



World's first solar highway (Credit: <http://chinaplus.cri.cn>)

Traditional logistics and transport systems have been improved by advancements in technologies such as artificial intelligence (AI). This article covers how AI has been transforming logistics and transport systems including self-driving vehicles, AI-robots in warehousing, AI-robots in healthcare and drones in logistics.

Impact of AI on logistics and supply chain

AI is going to have a huge impact on modern transport systems, which include movement of materials/goods, animals and humans. Logistics management involves planning and coordinating timely, safe and effective movement of products till they reach the destination, or the customers.

Irrespective of business location or industry, logistics plays an essential part in supply chain management, which deals with production, shipment, warehousing and delivery of products. Use of AI improves factory scheduling, production planning, boosts productivity and helps improve customer experience.

Supply chain is important to businesses because it directly affects sales and profits. But without effective and well-organised logistics, supply chain cannot help the business gain a clear advantage over its competition.

Logistics mainly depends on transportation of goods. Bullock carts and horses were replaced by trucks and railway wagons as an effective transport system improves the supply chain. Modern logistics and transport systems

include self-driving trucks, drones, robots and other means.

Self-driving trucks

Autonomous vehicles (AVs) are a major innovation of Industrial Revolution 4.0. In practice, it is easier to implement self-driving on trucks running on highways as compared to self-driving cars running in crowded marketplaces. Hence, commercial self-driving trucks have already been rolled out in parts of the US, Europe and Asia.

Trucks are one of the most common and important lifeline support systems in metros and cities. However, as per a US federal report, approximately 400,000 trucks crash each year, killing about 4000 people. Human error is the culprit in almost every accident.

One of the solutions is the use of AI-based self-driving trucks, which offer not only making roads safer but also cheaper and more efficient. One such example is Otto's self-driving truck that was acquired by Uber in

August 2016. Otto's system offers true level-four autonomy. In October 2016, an Otto truck achieved the longest continuous journey by a driverless and autonomous semi-truck, and travelled around 212.4km (132-mile) route from Colorado Springs to Fort Collins in Colorado, USA.

Another example is Vera, an autonomous, electric vehicle from Volvo.

Silicon Valley startup Plus.ai is said to be the first company to deploy a self-driving commercial truck to haul roughly 42 hours cross-country from Tulare, California, to Quakertown, Pennsylvania. The company's website claims it to be the best-in-class level-four autonomous driving system.

There is a lot of development in progress with AI technology in the logistics industry. "In the next five to ten years, 3.5 million truck drivers will be replaced by self-driving trucks," says Andrew Yang, author of *The War On Normal People*. The book talks about the problems caused by Industrial Revolution 4.0, automation including self-driving trucks, job displacement and proposed solutions.

Smart roads

Rise of self-driving vehicles and their specific criteria demands specific road infrastructure to support new and emerging technologies. Roadways have enormous surfaces stretching from one end to the other. If these huge surface areas could be used to generate electricity, the energy crisis could be overcome, especially for electric vehicles.

For example, SolaRoad project in the Netherlands is creating a seventy-metre stretch of bike path with integrated solar panels. Solar panels on the road will be used for lighting the road, traffic lights, road signs and



Plus.ai's first commercial self-driving truck (Credit: <https://plus.ai>)